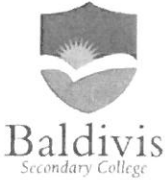


|                                                                                   |                                                                                                                                    |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Name: _____                                                                       | Date: _____                                                                                                                        |
| Teacher: _____                                                                    |                                                                                                                                    |
|  | <b>Year 9 Essentials – Investigation 1</b><br><b>Scientific Notation- Validation</b><br><b>Weighting %</b> _____ <b>Mark</b> _____ |

1. Complete the table

Use the information given to complete the table.

| Decimal form | Working                                 | Scientific notation    |
|--------------|-----------------------------------------|------------------------|
| 2 350 000    | $2.35 \times 1\,000\,000$               | $2.35 \times 10^6$     |
| 502 170      | $50217 \times 10000$                    | $5.0217 \times 10^5$   |
| 314 060 000  | $31406 \times 10\,000\,000$             | $3.1406 \times 10^8$   |
| 0.000298     | $2.98 \div 10\,000 = \frac{2.98}{10^4}$ | $2.98 \times 10^{-4}$  |
| 0.000004621  | $4621 \div 10^6$                        | $4.621 \times 10^{-6}$ |
| 0.003082     | $3082 \div 10^3$                        | $3.082 \times 10^{-3}$ |

What do you notice about scientific notation form? How do you know it is correctly in scientific notation?

always 1 digit in front of decimal point

2. Complete the table.

| Scientific Notation    | Working               | Decimal Form   |
|------------------------|-----------------------|----------------|
| $5.34 \times 10^7$     | <u>5.3400000</u>      | 53400000       |
| $7.4 \times 10^5$      | <u>7.40000</u>        | 740000         |
| $4.25 \times 10^{-6}$  | <u>0.00000425</u>     | 0.00000425     |
| $3.02 \times 10^{-10}$ | <u>0.000000000302</u> | 0.000000000302 |

3. Put the following in ascending order (smallest to largest)

a)  $8 \times 10^5$ ,  $8 \times 10^4$ ,  $8 \times 10^3$

$8 \times 10^3$ ,  $8 \times 10^4$ ,  $8 \times 10^5$

b)  $5.3 \times 10^{-3}$ ,  $6.2 \times 10^{-3}$ ,  $8 \times 10^{-6}$

$8 \times 10^{-6}$ ,  $5.3 \times 10^{-3}$ ,  $6.2 \times 10^{-3}$

c) 6300,  $3.4 \times 10^4$ ,  $8 \times 10^3$

$6.3 \times 10^3$

6300,  $8 \times 10^3$ ,  $3.4 \times 10^4$

When ordering numbers in scientific how do you know one number is larger than the other?

higher power

4. Simplify the following expressing your answers in scientific notation

a)  $(8 \times 10^4)^2$

$$8^2 \times 10^8 = 64 \times 10^8 = 6.4 \times 10^9$$

b)  $(3.4 \times 10^4) \times (2 \times 10^3)$

$$3.4 \times 10^4 \times 2 \times 10^3 = 6.8 \times 10^7$$

c)  $(9.3 \times 10^6) \div (3.1 \times 10^4)$

$$\frac{9.3 \times 10^6}{3.1 \times 10^4} = 3 \times 10^2$$

5. Which of the following is the correct way to represent 470 000 in scientific notation?

a)  $47 \times 10^4$

b)  $4.7 \times 10^5$

6. In question 5, one of the previous answer is correct. What is wrong with the incorrect answer? Why is it not in scientific notation?

~~47~~ has more than one digit in front of decimal point

